Day 38 EBIKE101 Instructor Packet

CTS Cohort 6 | EBIKE101 | Tuesday 2026-09-08

Day 38 EBIKE101 Instructor Run Sheet

Program: CTS Cohort 6

Course: EBIKE101

Week: Final week of CTS / Week 2 of EBIKE101

Date: Tuesday, 2026-09-08

Context: Monday 2026-09-07 was a holiday. Resume with a practical diagnostic day.

Daily focus: Diagnose the E-bike Homunculous, identify what is needed to finish it, and begin safe finish work.

Big idea

A project is not “almost done” because it looks like a bike. It is close to done when the team can explain what is installed, what is missing, what is verified, what is unsafe, and what the next safe work step is.

Student-facing goal

By the end of today, students should be able to:

1. Inventory an unfinished e-bike build.
2. Separate mechanical problems from electrical/control problems.
3. Identify missing parts or unknowns.
4. Trace source, protection, control, and load paths.
5. Create a finish-work list that is safe, ordered, and realistic.
6. Begin work only on tasks that are safe and approved.

Materials

- E-bike Homunculous project
- Basic hand tools
- PPE
- Multimeters
- Zip ties / wire management materials
- Labels / tape / marker
- Notebook or printed diagnostic worksheet
- Camera or phone for evidence photos
- Known-good charger/battery info if available
- Any available controller/display/PAS/throttle documentation

Safety boundary

Students may inspect, document, trace, label, and perform approved mechanical/wire-management work. Battery connection, powered testing, controller testing, and motor testing require instructor control.

Do not allow:

- Battery shorting experiments
- Continuity testing on live battery circuits
- Bypassing fuses
- Full-throttle tests
- Ride testing before mechanical and electrical checks
- Hands near chain, spokes, wheel, or drivetrain during powered testing

Schedule

10:00–10:15 — Arrival / reset / context

- Remind students Monday was a holiday and today restarts the build thread.
- Name the project: E-bike Homunculous diagnosis and finish work.
- Frame the day as “find out what this bike needs,” not “make it work at any cost.”

Instructor line:

“We are not trying to prove the bike works. We are trying to prove what is true about it.”

10:15–10:35 — Morning quiz

- Give student quiz.
- Collect or review immediately.
- Use missed questions to decide reteach focus.

10:35–10:55 — Safety and system map mini-lesson

Cover:

- Source: battery/charger
- Protection: fuse/breaker/BMS/controller limits
- Control: display, PAS, throttle, brake cutoff, key/switch
- Load: motor, lights, accessories
- Frame/mechanical: brakes, wheels, torque arms, chain, mounts

Key point:

A silent motor does not prove a bad motor. It can be missing power, missing control input, active brake cutoff, bad connector, wrong display/controller pairing, bad PAS, blown fuse, or mechanical lockup.

10:55–11:30 — Whole-bike inventory

Students work in pairs or one full group depending on class size.

Inventory categories:

- Frame and mounting
- Wheels/tires/brakes
- Battery and mount
- Controller and mount
- Display/control parts

- PAS/throttle/brake cutoffs
- Motor wiring
- Connectors
- Protection/fusing
- Missing parts
- Unknowns

11:30–12:00 — Diagnostic board

Create a board with four columns:

1. Verified
2. Unknown
3. Missing/Needed
4. Unsafe until fixed

Students must support every claim with observation, measurement, or documented uncertainty.

12:00–12:30 — Lunch / reset

Before break, make the bike safe:

- Battery disconnected or controlled by instructor
- Tools reset
- Loose parts contained
- Wires not left under tension

12:30–1:00 — Finish-work plan

As a class, choose approved tasks:

- Mechanical tightening/inspection
- Brake adjustment
- Cable routing
- Labeling
- Connector inspection
- Mounting cleanup
- Parts list
- Documentation/photos
- Non-powered continuity checks only where appropriate

Do not jump straight to powered testing because impatience is not an electrical method.

1:00–2:20 — Approved work block

Students rotate through roles:

- Lead mechanic: physical fitment, fasteners, brakes
- Electrical tracer: source/control/load path map
- Scribe: worksheet, parts list, evidence log
- Safety observer: pinch points, sharp edges, battery boundary
- Parts runner: organizes missing/needed components

Instructor circulates and signs off before any higher-risk work.

2:20–2:45 — Controlled test decision

Do not guarantee powered testing. Decide from evidence.

Powered test may happen only if:

- Wheel is secure
- Brakes function
- Axle/torque reaction is controlled
- Battery voltage/rating is appropriate
- Fuse/protection path is present
- No damaged/melted wiring
- Connectors are seated and strain-relieved
- Drive wheel is safely lifted or restrained
- Instructor controls first power-up

2:45–3:00 — Closeout

Students complete:

- What we verified
- What is still unknown
- Parts needed
- Next safe action
- Photo evidence list

Instructor closeout notes

Today is successful if the class leaves with a clear diagnostic state, not necessarily a finished bike. That distinction will save you from the ancient human urge to call chaos “progress” because something briefly spun.

Day 38 EBIKE101 Morning Quiz — Instructor Answer Key

Program: CTS Cohort 6

Course: EBIKE101

Date: Tuesday, 2026-09-08

Topic: E-bike Homunculous diagnosis and finish-work planning

Answer Key with Rationales

1. **B** — Diagnosis starts with safety. Students should confirm the system is safe to inspect and de-energized when appropriate before touching unknown wiring.

2. **A** — An unfinished build needs inventory first: what exists, what is missing, what is loose, what is unknown, and what must be verified.

3. **B** — A fuse or breaker protects wiring and parts from unsafe current. It is not a performance part.

4. **B** — A dead display can come from missing battery voltage, switch/fuse issues, or a disconnected display/controller path.

5. **B** — A missing throttle is a missing control input. It does not automatically prove the system is impossible, but it must be documented and checked against the controller/display/PAS behavior.

6. **B** — Wire color can be a clue, but e-bike systems often involve adapters, repairs, and mixed manufacturers. Testing beats color-confidence theater.

7. **A** — Continuity testing belongs on a de-energized circuit or component. Using continuity mode on a live battery circuit can damage the meter or create unsafe conditions.

8. **A** — Brake cutoffs tell the controller not to drive the motor. A stuck or misadjusted cutoff can make a working system appear dead.

9. **A** — PAS usually means Pedal Assist Sensor/System. It detects pedaling input so the controller can assist.

10. **B** — Strain relief protects wires from pulling, rubbing, bending, and wheel movement.

11. **B** — A blown fuse is a symptom. Before replacement, inspect for shorts, wrong connections, pinched wires, or overload.

12. **B** — Tracing means following the real path from source, through protection/control, to load.

13. **B** — Powered testing requires mechanical safety first: secure wheel, axle/torque support, brakes, chain path, and clear moving parts.

14. **B** — First powered testing should be controlled, low-power, secured, and instructor-led. No shop test rides.

15. **B** — A useful diagnostic log captures symptoms, observations, measurements, changes, parts needed, and next step.

16. **A** — The controller turns rider/display inputs into controlled electrical power for the motor.

17. **B** — Heat, burnt smell, or melted wiring means stop and get instructor review. More testing can make damage or fire risk worse.

18. **A** — Connectors fail in boring ways: bent pins, loose plugs, corrosion, bad matches, or damaged wires. Naturally, boring failures create dramatic symptoms.

19. **B** — A symptom is what is observed. A root cause is the underlying reason.

20. **B** — “Finished” means verified mechanical safety, electrical safety, control function, supported wiring, known missing parts, and documentation.

Suggested grading

- 18–20: Ready for diagnostic leadership role today.
- 15–17: Solid working understanding; pair with peer verification.
- 12–14: Review safety and system tracing before powered work.
- Below 12: Keep student in observation/documentation/mechanical-support role until reteach.

Reteach focus

If many students miss:

- Questions 1, 7, 14, 17: reteach safety boundaries before lab.
- Questions 5, 8, 9, 16: reteach control inputs and controller logic.
- Questions 11, 12, 15, 19: reteach troubleshooting process and documentation.

Day 38 EBIKE101 Diagnostic Worksheet

Program: CTS Cohort 6

Course: EBIKE101

Date: Tuesday, 2026-09-08

Project: E-bike Homunculous

Team members: _____

Purpose

Use this worksheet to diagnose what the Homunculous needs before it can be safely finished, powered, and eventually ridden.

Safety boundary

Do not connect the battery, bypass protection, open battery cases, test live battery circuits in continuity mode, spin the motor under power, or ride the bike without instructor approval.

1. Whole-bike inventory

2. System trace

Fill this out before powered testing.

3. Symptom list

4. Missing parts / supplies list

5. Safe work approved today

Only list work the instructor has approved.

6. Evidence log

7. Exit reflection

What is one thing we verified today?

_____

What is one thing we still do not know?

_____

What is the next safest work step?

_____

Day 38 EBIKE101 Afternoon Lab — Homunculous Finish-Work Diagnosis

Program: CTS Cohort 6

Course: EBIKE101

Date: Tuesday, 2026-09-08

Lab type: Live project diagnosis and approved finish work

Lab objective

Students will diagnose the current state of the E-bike Homunculous, identify what is required to finish it, and begin only the safe work that has been verified and approved.

Instructor setup

Before students begin:

1. Place the bike where all sides can be accessed.
2. Keep battery connection under instructor control.
3. Remove or isolate any obviously unsafe condition.
4. Stage hand tools, meter, labels, zip ties, and worksheet.
5. Define the boundary: diagnosis first, power later only if earned by evidence.

Lab roles

Assign or rotate:

- Mechanical lead: checks frame, wheels, brakes, fasteners, chain path.
- Electrical tracer: maps source, protection, controller, controls, motor.
- Documentation lead: worksheet, board, photo evidence, parts list.
- Safety observer: watches pinch points, battery boundary, tool use, moving parts.
- Parts lead: organizes missing parts, hardware, adapters, labels.

Part 1 — Mechanical readiness inspection

Students inspect:

- Wheel security
- Axle nut condition
- Torque arm or anti-rotation support if applicable
- Brake function
- Tire condition and pressure
- Chain alignment
- Crank/pedal condition
- Sharp edges
- Cable rub points
- Loose fasteners
- Battery/controller mounting

Stop if:

- Motor wheel can shift in dropout

- Brakes cannot stop the wheel
- Chain path is unsafe
- Battery mount is loose
- Any wire can reach tire, chain, brake rotor, or crank

Part 2 — Electrical/control inventory

Students identify and document:

- Battery voltage/rating if labeled
- Controller label/rating if visible
- Main power connector
- Fuse/protection location
- Display connector
- PAS connector/sensor/magnet ring
- Throttle presence or absence
- Brake cutoff connectors
- Motor cable
- Accessory wires
- Damaged connectors
- Unknown wires

Rule:

Unknown wires stay unknown until traced, labeled, or verified. Guessing does not become truth just because someone says it confidently in a shop voice.

Part 3 — Diagnostic decision tree

Use this order:

1. Is the bike mechanically safe to inspect?
2. Is the battery appropriate and controlled?
3. Is there protection between battery and controller?
4. Does the controller have a secure mount?
5. Are control inputs present?
6. Are brake cutoffs present and not stuck active?
7. Is the PAS sensor installed and aligned?
8. Is the throttle present, missing, or optional for this controller?
9. Is the motor cable undamaged and supported?
10. What must be fixed before first power-up?

Part 4 — Approved finish work

Allowed student work may include:

- Brake adjustment
- Tightening non-critical hardware under supervision

- Labeling wires
- Improving cable routing
- Adding strain relief
- Cleaning mounting surfaces
- Organizing connectors
- Creating a parts-needed list
- Documenting evidence
- Non-powered meter checks where appropriate

Instructor-only or instructor-controlled work:

- Battery connection
- First power-up
- Controller testing
- Motor spin testing
- Fuse replacement if cause is unknown
- Any modification to high-current wiring
- Any ride test

Part 5 — First power-up gate

Do not perform a powered test unless all are checked:

- Mechanical assembly secure
- Brakes functional
- Drive wheel controlled
- Battery voltage/rating appropriate
- Fuse/protection path verified
- Controller mounted
- No melted/damaged wires
- No unrestrained motor cables
- No hands near moving parts
- Instructor gives explicit approval

Completion target

The lab is complete when the team can answer:

1. What works or appears complete?
2. What is missing?
3. What is unsafe?
4. What is unknown?
5. What should be done next?

Finishing the bike is excellent. Finishing the diagnosis is required.